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Where Minds Pollinate

Physics
Class 11

FBISE

Chapter 4
Handwritten Notes

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Day: Tuesday Date: 15/09/2026

Unit 04: "Rotational And Circular motion"

4.1: Rotational Kinematics

1. Rotational motion:

"Rotational motion is the turning or spinning motion of an object about a fixed axis."

- **Main Idea:**
  object rotates → around an axis
  → different points move in circles.

➡ Rotation axis: the line about which the object rotates.

➡ The axis remains fixed while the points of the object move in circular path.

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- Examples:-

- Fan rotating about its central axis.
- wheel rotating about its axle.
- Earth rotating about its axis.

Important point :-

The rotation axis may be a pivot, hinge or any fixed support.

❖ Rotational kinematics:-

- Rotational kinematics deals with the motion of objects along a circular path without considering forces or torques.

It mainly describes:

- Angular position
- Angular displacement
- Angular velocity
- Angular acceleration

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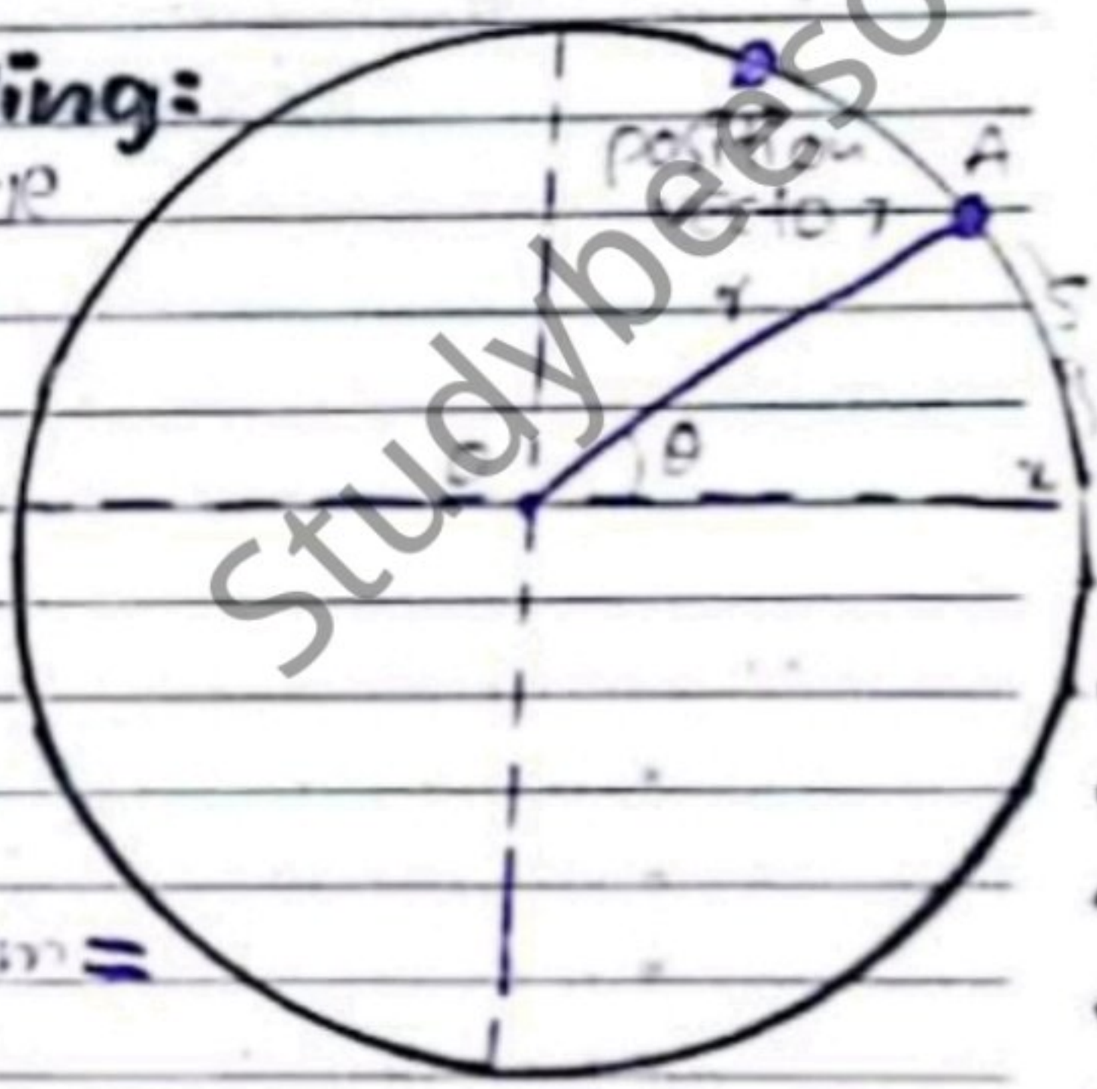
Angular Position:-

- **Definition:-**

"The angle through which the position vector of a moving object is displaced from a chosen reference direction is called angular position (θ)"

- **Easy Understanding:**

Consider a particle moving around a circle.



Reference line

↓

Position of particle

↓

The angle b/w them = Angular position

- **Formula:**

$\theta = \frac{S}{r}$ ◻ Where;

θ = Angular position
S = arc length
r = radius of circle

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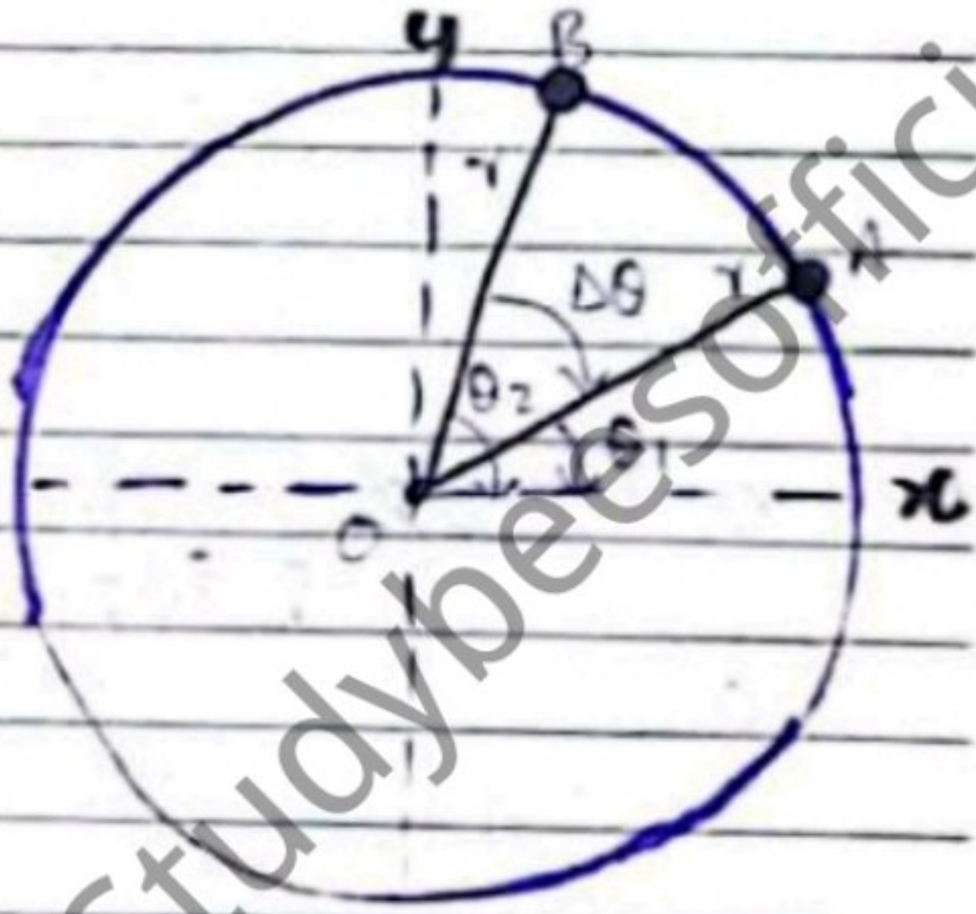
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Angular displacement :-

—o Δθ o—

- **Definition:**

"The change in angular position of an object is called angular displacement."



Initial position (A)
↓ Rotation
final position (B)

- **Formula:**

$$\Delta\theta = \theta_f - \theta_i$$

- where;

θ_f = initial angular position
θ_i = final angular position
$\Delta\theta$ = angular displacement

- **Remember:**

Change in angular position
↓
Angular displacement

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❖ Direction of Angular displacement:-

- Angular displacement is a vector, so it has:
  - ➡ Magnitude
  - ➡ Direction

- **Sign convention:**

Anticlockwise rotation → Positive (+)

Clockwise rotation → Negative (-)

❖ Units of Angular Displacement:-

- The SI unit of angular displacement is:
  - radian (rad).
- Other units:
  - Degree (°)
  - Revolution (rev).

❖ Radian and Degree:

- One complete revolution:-
  - 1 rev = 360°.
- and
  - 1 rev = 2π rad

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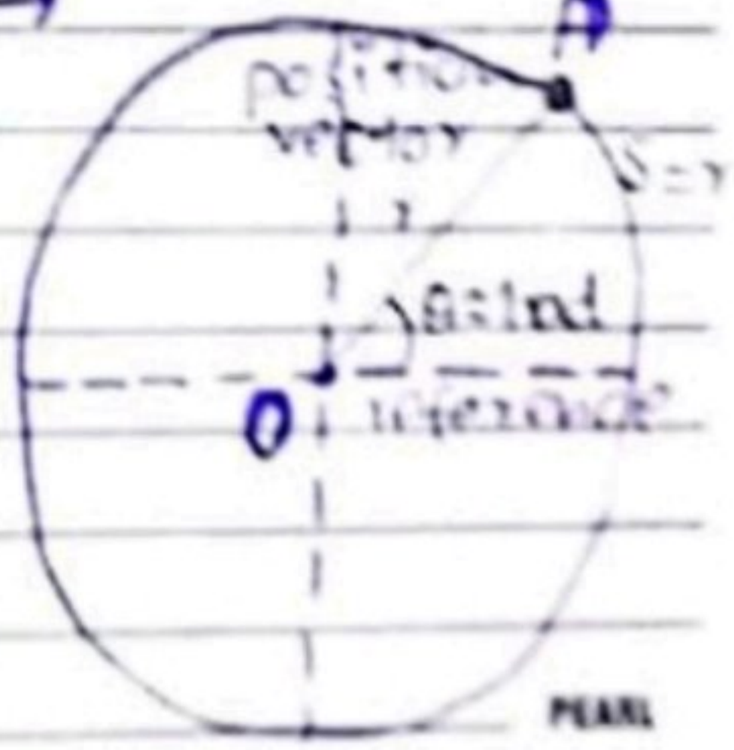
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- Therefore:
  $2\pi rad = 360°$
- Most important conversion:-
  $1 rad = \frac{180}{\pi} = 57.3°$
- Quick conversion:-
- Degrees → Radians
  $\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$
- Radians → Degrees
  $\theta_{deg} = \theta_{rad} \times \frac{180}{\pi}$

- **What is 1 radian?**

"1 radian is the angle subtended at the centre of a circle by an arc whose length is equal to the radius of the circle."

- Easy Memory trick:-
  Arc length = Radius → Angle = 1 radian

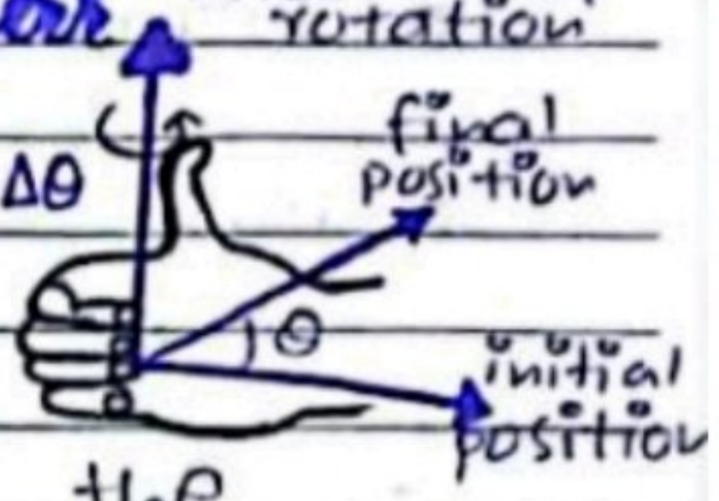


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❖ Direction of Angular Displacement — Right hand rule :—



- The Right Hand rule is used to find the direction of angular displacement.

- How to use it?
  - Curl the fingers of your right hand in the direction of rotation
  - Your thumb points in the direction of angular displacement.

- Remember:
  - Fingers → Rotation direction
  - Thumb → Angular displacement direction.

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Angular Velocity:-

-(ω)-

- **Definition:**

"The time rate of change of angular displacement is called angular velocity."

- In simple words:-
  Angular displacement changes → with time → Angular velocity

- **Formula:**

$$\omega = \frac{\Delta\theta}{\Delta t}$$

- Where;
  - ω = angular velocity
  - $\Delta\theta$ = angular displacement
  - Δt = time interval

Units of Angular Velocity:-

- SI unit:
  $rad\ s^{-1}$
- Other units:
  - degree /s
  - revolution /s
  - revolution / minute (rpm)

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- **Direction:**

"The direction of angular velocity is the same as the direction of angular displacement"

Therefore:

- Anticlockwise → $+\omega$
- Clockwise → $-\omega$

Average Angular velocity:-

- **Definition:**

"The total angular displacement during a given interval of time divided by that time is called average angular velocity"

- **Formula:**

$$\omega_{av} = \frac{\theta}{t} \quad \text{or} \quad \omega_{av} = \frac{\Delta\theta}{\Delta t}$$

Instantaneous Angular velocity:-

- **Definition:**

"The angular velocity of an object at a particular instant of time is called instataneous angular velocity."

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- It is obtained when the time interval becomes extremely small:

$$\omega_{inst} = \lim_{\Delta t \to 0} \frac{\Delta \theta}{\Delta t}$$

Angular Acceleration:-

-(α)-

- **Definition:**

"The time rate of change of angular velocity is called angular acceleration."

- **Formula:**

$$\alpha = \frac{\Delta w}{\Delta t}$$

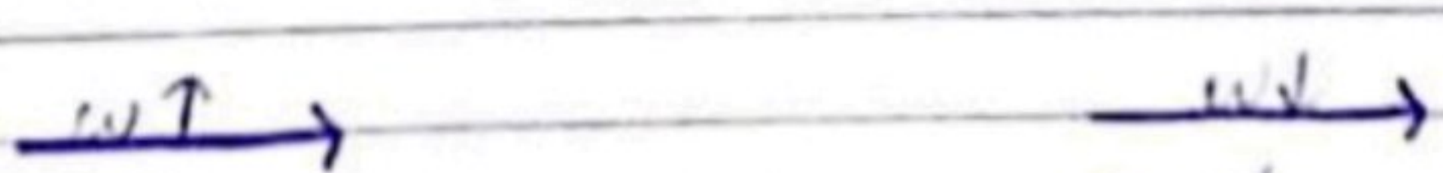
- where;
  - α = angular acceleration
  - Δw = angular velocity
  - Δt = change in time.

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Units of Angular acceleration:

- The SI unit of angular acceration is
  rad/s^2
- Others:-
  - deg/s^2
  - rev/s^2
- Direction of angular acceleration is determined by right hand rule
- Angular $\vec{\alpha}$ is taken positive when:-
  - angular velocity increases.
  - In such case angular velocity and angular $\vec{\alpha}$ have same direction.
- α is taken negative when:-
  - angular velocity decreases.
  - In such case angular $\vec{\omega}$ and angular $\vec{\alpha}$ are antiparallel.



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❖ Average Angular acceleration:

"The total angular velocity 'ω' of a body in time 't' is called angular acceleration."

Formula:

$$\alpha_{avg} = \frac{\omega}{t}$$

❖ Instantaneous Average acceleration:-

"The limiting value of the ratio between Small change in angular velocity 'ω' and small time interval 't', such that the time approaches to zero, is called instantaneous velocity."

Formula:

$$\alpha_{inst} = \lim_{\Delta t \to 0} \frac{\Delta \omega}{\Delta t}$$

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Relationship b/w Linear & Angular Kinematic Quantities:-

1. Linear ↔ Angular Displacement:-

- When a particle moves from A to B along a circular path:
  - Arc length = linear displacement S
  - Angle turned at the centre = angular displacement θ.
  - Radius of circle = r

- For 1 radian:
  - Arc length = r

- Therefore, for any angular displacement measured in radians:

$$S = r\theta$$

- Remember!
  - S = linear displacement (m)
  - r = radius (m)
  - θ = angular position (rad).

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- Key relationship:

$$\theta = \frac{S}{r}$$

- Concepts:
  - If the radius is larger, the same angular rotation produces a larger arc length.

2. linear ↔ Angular velocity:-

- Starting from:

$$S = r\theta$$

- Differentiate w.r.t time:

$$v = r\omega$$

- Where:

  v = linear velocity
  r = radius
  ω = angular velocity

- Tangential velocity:-
  - The instantaneous linear velocity of an object moving in a circle is directed along the tangent to the circular path.

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- Therefore:

$v = r\omega$

It is also called tangential velocity relation.

- Concept Check:
- If ω remains constant:

$v \propto r$

- So, a point farther from the centre has greater linear / tangential velocity.

3. Linear ↔ Angular Acceleration:

- In circular motion, acceleration generally has two components;

$$\vec{a} = \vec{a}_T + \vec{a}_R$$

- Where;
  - a_T = tangential acceleration
  - a_R = radial acceleration.
- Because these components are perpendicular:

$$a = \sqrt{a_T^2 + a_R^2}$$

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A. Tangential Acceleration:-

- Tangential acceleration is "the component of acceleration parallel to the instantaneous linear velocity."

- What causes it?
  It occurs when the magnitude of velocity changes.

- From:
  $v = r\omega$
- we obtain:
  $a_T = r\alpha$
- where:
  a_T = tangential acceleration
  r = radius
  α = angular acceleration

B. Radial Acceleration:-

- Radial acceleration acts along the radius, toward the centre of the circular path.

- What causes it?
  It occurs when the direction of velocity changes.

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- For any object moving in a circle with constant speed, there is still acceleration b/c the direction of velocity continuously changes.

- This acceleration is also called:
  "centripetal acceleration"
- and

$$a_R = \frac{v^2}{r}$$

- Since $v = r\omega$:

$$a_R = \frac{(r\omega)^2}{r} \Rightarrow a_R = \frac{r^2\omega^2}{r}$$

$$a_R = r\omega^2$$

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Centripetal acceleration & Centripetal force :-

What is circular motion?

- When an object moves along a circular path, its direction of velocity continuously changes, even if its speed remain constant.
- Therefore the object is accelerating
- This acceleration is called "centripetal acceleration"

Definition:

"Centripetal acceleration is the acceleration of an object moving in a circular path, directed towards the centre of the circle."

- It is represented by:

$$a_c = \frac{v^2}{r} \quad \text{or} \quad a_c = r\omega^2$$

What is centripetal force?

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- According to newton's second law, acceleration requires a net force.
- Therefore, the inward acceleration of a body in circular motion requires an inward net force.

Definition - Centripetal force:

"The net force that causes a particle to undergo centripetal acceleration is called centripetal force."

$$F_c = ma_c$$

Therefore;

$$F_c = \frac{mv^2}{r}$$

and;

$$F_c = mr\omega^2$$

Basic Concepts :-

- **Uniform circular motion:**

Motion in a circle + constant speed

- Although speed is constant:
  - velocity is changing b/c its speed direction changes.

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- therefore acceleration exists.
- this acceleration points toward the centre.

- Example:~
A stone tied to a string and rotated in a circle has approximately uniform circular motion it its speed remains constant.

Velocity in circular motion?

At every point on a circular path, velocity is tangent to the circle. ∴

So:

$\vec{v} \perp \vec{r}$



~~Even if:~~

~~$|\vec{v}_i| = |\vec{v}_f|$~~

~~their directions are different.~~

~~Therefore:~~

~~$\Delta\vec{v} = \vec{v}_f - \vec{v}_i$~~

~~and:~~

Why does centripetal acceleration occur?

Suppose a particle moves from point A to B.

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- its velocity changes from $\vec{v}_i$ to $\vec{v}_f$.
- Even if: $|\vec{v}_i| = |\vec{v}_f|$
- their directions are different.

Therefore:

$$\Delta\vec{v} = \vec{v}_f - \vec{v}_i$$

- and

$$\vec{a} = \frac{\Delta\vec{v}}{\Delta t}$$

- As the time interval becomes extremely small, the acceleration points toward the centre.

Direction of centripetal acceleration:

But veto

"$\vec{a}_c$ is always directed towards the centre"

- But velocity is tangent to the circle.
- Therefore:

$$\vec{a}_c \perp \vec{v}$$

- **Imp mcqs point:-**

velocity → tangent

centripetal acceleration → centre

centripetal force → centre.

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❖ Centripetal force is not a new type of force?

- "Centripetal force" simply means:
  - The net inward force responsible for circular motion.
- It may be provided by:

Situation:	Force providing centripetal force:
Stone tied to String	Tension
Car on a flat road	Friction
Planet around Sun	Gravitational
Satallite around Earth	Gravitational
Car on a banked road	Normal
Charged particle in magnetic field.	Magetic

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What happens if centripetal force disappears?

- If the inward force suddenly disappears, the object doesn't move directly away from the centre.
- It moves along the tangent to the circle at that point.

- **Reason:**
- At that instant, the velocity is tangent to the circle.

Centripetal force and speed:

- From:

$$F_c = \frac{mv^2}{r}$$

- we see:

$$F_c \propto v^2$$

- Therefore, if speed doubles:

$$F'_c = \frac{m}{r}(2v)^2$$

$$F'_c = 4F_c$$

- So doubling speed makes centripetal force **four times.**

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Effect of radius:-

$$F_c = \frac{mv^2}{r}$$

- Thus;

$$F_c \propto \frac{1}{r}$$

- For constant m & v:~
  - larger. $\uparrow r \rightarrow \downarrow F$
  - $\downarrow r \rightarrow \uparrow F$

SI unit: ms^{-2}

Dimension: $[\ ]T^{-2}]$

Formulae & Units:-

Formula 01: Centripetal acceleration

$$a_c = \frac{v^2}{r}$$

Where:

a_c = centripetal : acceleration

v = linear speed

r = radius

SI

Formula 02: Angular form

Since:

$$v = r\omega$$

then:

$$a_c = \frac{(r\omega)^2}{r}$$

$$a_c = r\omega^2$$

SI unit:

ms^{-2}

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Formula 03: Centripetal force

$$F_c = m a_c$$

Therefore;

$$F_c = \frac{mv^2}{r}$$

SI unit:
Newton (N)

Dimensions:
$[MLT^{-2}]$

Formula 04: Angular form of force

$$F_c = m\omega^2 r$$

This is one of the formula specially mentioned in the FBISE SLO.

Formula 05: liner - angular velocity relation

$$v = r\omega$$

where;
v = linear velocity (m/s)
r = radius (m)
ω = angular velocity (rad/s).

Formula 06: Angular velocity

$$\omega = \frac{\theta}{t}$$

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- For one complete revolution:

$$\theta = 2\pi$$

- Therefore:

$$\omega = \frac{2\pi}{T}$$

∴

- where T is the time period:

Also:

$$\boxed{\omega = 2\pi f} \qquad \because f = \frac{1}{T}$$

Formula 07: Banked road

For an ideal frictionless banked road:

$$\tan\theta = \frac{v^2}{rg}$$

Therefore:

$$v = \sqrt{rg\tan\theta}$$

Formula 08: Maximum speed on an unbanked road

If friction provides the centripetal force:

$$f_s = \frac{mv^2}{r}$$

At limiting condition:

$$f_{s\,max} = \mu_s N$$

For a horizontal road:

$$N = mg$$

Therefore:

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$$v_{max} = \sqrt{\mu_s r g}$$

Derrivation Of Centripetal Acceleration Force:

- Consider a particle moving uniformly in a circle.
- At A:

$\vec{v}_i$

- At B:

$\vec{v}_f$

- Since the motion is uniform:

$$|\vec{v}_i| = |\vec{v}_f| = v$$

- The change in velocity is:

$$\Delta\vec{v} = \vec{v}_f - \vec{v}_i$$

- From the similar triangles formed by the velocity vectors and radius vectors:

$$\frac{\Delta v}{v} = \frac{\Delta r}{r}$$

- Therefore:

$$\Delta v = \frac{v}{r} \Delta r$$

- Divide by Δt:

$$\frac{\Delta v}{\Delta t} = \frac{v}{r} \cdot \frac{\Delta r}{\Delta t}$$

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- But;

$$\frac{\Delta v}{\Delta t} = a$$

- and in limiting case:

$$\lim_{\Delta t \to 0} \frac{\Delta r}{\Delta t} = v$$

- Therefore:

$$a_c = \frac{v}{r} v$$

- Hence:

$$a_c = \frac{v^2}{r}$$

Physical meaning:

The acceleration is necessary to continuously change the direction of velocity & keep the object on a circular path

Derivation Of $a_c = r\omega^2$

- we know:

$$v = r\omega$$

- and:

$$a_c = \frac{v^2}{r}$$

- Substitute:

$$a_c = \frac{(r\omega)^2}{r}$$

$$a_c = \frac{r^2\omega^2}{r}$$

- Therefore:

$$a_c = r\omega^2$$

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1) Derivation of Centripetal force:

Newtons second law:

$$F = ma$$

for circular motion:

$$a = a_c$$

Therefore:

$$F_c = ma_c$$

using:

$$a_c = \frac{v^2}{r}$$

we get:

$$F_c = m \cdot \frac{v^2}{r}$$

Hence:

$$F_c = \frac{mv^2}{r}$$

Using:

$$v = r\omega$$

we get:

$$F_c = \frac{mr^2\omega^2}{r}$$

Hence:

$$\boxed{F_c = mr\omega^2}$$

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Friction as centripetal force:

- For a car on unbanked circular road, static friction can provide the required inward force.

$$f_s = \frac{mv^2}{r}$$

- At maximum speed:

$$f_{s\,max} = \mu_s N$$

- Since:

$$N = mg$$

$$\mu_s mg = \frac{mv^2}{r}$$

- Cancel m:

$$\mu_s g = \frac{v^2}{r}$$

- Therefore:

$$v_{max} = \sqrt{\mu_s r g}$$

Important conclusion:

- The mass cancels. So, under this idealized model, maximum speed doesn't depend on the mass of the car.

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Tension as centripetal force:-

- Suppose a ball of mass 'm' is attached to a string and moves in a circle.
- The Tension points towards the centre.
- Therefore:

$$T = F_c$$

$$T = \frac{mv^2}{r}$$

Gravitational force as centripetal force:-

- For a satellite orbiting Earth, gravity provides the centripetal force.

$$F_g = F_c$$

- Therefore:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

- Cancel 'm':

$$\frac{GM}{r} = v^2$$

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- Thus:

$$v = \sqrt{\frac{GM}{r}}$$

"This shows that orbital motion is maintained by gravitational attraction."

Why do satellites appear weightless?

- A satallite and everything inside it are continuously falling towards Earth under gravity.
- At the same time, the satallite has sufficient tangential velocity to keep missing Earth's surface and remain in orbit.
- The objects inside are therefore in free fall (continuous).
- Consequently, they do not exert the usual supporting force on the floor of the spacecraft so they appear weightless.

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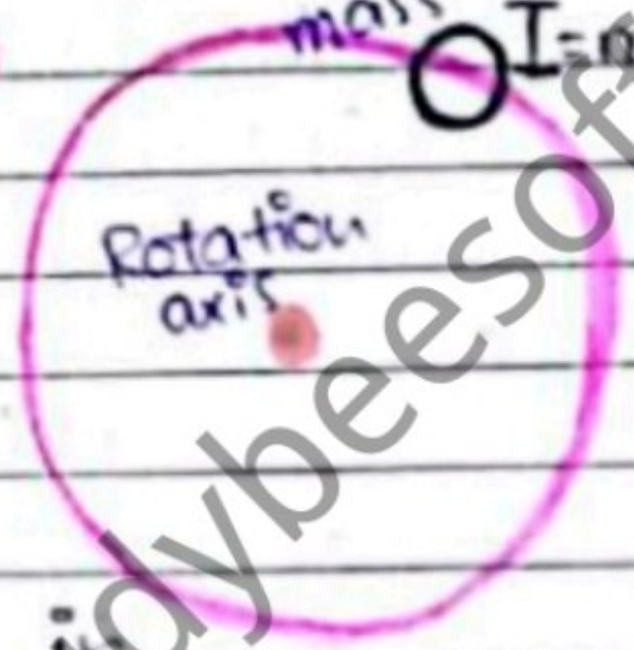
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Moment Of Inertia:=

Definition:

"Moment of inertia (I) is the property of a body due to which it tends to maintain its state of rest or uniform rotational motion about a fixed axis."

It is also called: "rotational axis"



Import Concept:

- Moment of inertia rotational motion is similar to mass in linear motion.
  - Greater mass → greater inertia
  - Greater moment → greater resistance of inertia to angular a

For Point mass:

- If a mass **m** is rotating at a perpendicular distance **r** from the axis of rotatio

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$$I = mr^2$$

- **SI Unit:** kgm^2
- **Dimensions:** $[ML^2]$
- **Important points:**
  - Moment of inertia depends on:
    1. Mass of the body.
    2. Distribution of mass about the axis.
    3. Position of the axis of rotation.

> "The farther the mass is from the axis, the greator the moment of inertia."

Moment of Inertia of a Rigid Body:-

- A rigid body can be considered as a large no. of small masses:

$$m_1, m_2, m_3, \cdots, m_n$$

- at distances:

$$r_1, r_2, r_3, \cdots, r_n$$

from the axis.

Therefore;

$$I = \sum_{i=1}^{n} m_i r_i^2$$

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$$I = m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + \dots + m_nr_n^2$$

Remember!

Mass farther from the axis contributes more strongly because distance is squared.

Moment of inertia of common uniform objects:-

Object:	Moment of inertia:
Thin cylinder / rod about centre	$I = \frac{1}{12} ML^2$
Cylinder about end	$I = \frac{1}{3} MR^2$
Rectangular plate about centre	$I = \frac{1}{12} M(a^2 + b^2)$
Rectangular plate about edge	$I = \frac{1}{3} M(a^2 + b^2)$
Solid cylinder / disc	$I = \frac{1}{2} MR^2$
Ring / hoop	$I = \frac{1}{1} MR^2$

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Hollow cylinder	$I = \frac{1}{2} M(R_1^2 + R_2^2)$
Solid sphere	$I = \frac{2}{5} MR^2$
Hollow sphere	$I = \frac{2}{3} MR^2$

- Always look carefully at where the rotation axis is located before sketching the formula. -

Angular Momentum:-

Definition:

"Angular momentum $\vec{L}$ of an object is defined as the cross product of its position vector with its linear momentum."

$$\vec{L} = \vec{r} \times \vec{p}$$

Magnitude:

$$L = rp \sin\theta \hat{n}$$

where:

r = perpendicular distance from axis.

p = linear momentum.

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- θ = Angle b/w $\vec{r}$ & $\vec{p}$
- $\hat{n}$ = position vector

- For circular motion:

$$\theta = 90°$$



and

$$\sin 90° = 1$$

Therefore:

$$L = rp$$

SI Unit Of Angular momentum:

Since:

$$p = mv \quad \&$$

$$L = rp$$

Therefore SI unit:

$$kgm^2s^{-1}$$

Dimensions:

$$[M^1L^2T^{-1}]$$

Angular momentum of a point mass

- Consider a mass **m** rotating at distance: **r**.
- Step 1: Definition

$$L = rp$$

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- Step 2: Linear Momentum

$$p = mv$$

Therefore:

$$L = rmv$$

- Step 3: Relation b/w linear and angular velocity

$$v = r\omega$$

- After puting:

$$L = mr^2\omega$$

- But:

$$I = mr^2$$

- Therefore:

$$\mathbf{L = I\omega}$$

Angular momentum = Moment of inertia x Angular velocity

COMPARISON	
Linear	**Rotational**
o Mass (m)	o moment of inertia (I)
o Velocity (v)	o Angular velocity (ω)
o Linear momentum (p)	o Angular momentum (L)

Angular momentum of a rigid body

A rigid body consists of many small masses, $m_1, m_2, m_3, \ldots, m_n$

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at distances

$r_1, r_2, r_3, \cdots, r_n$

from the axis.

- The total anglar momentum is:

$$L_{net} = L_1 + L_2 + L_3 + \cdots + L_n.$$

- For each mass:

$$L_1 = m_1 r_1^2 \omega_1$$
$$L_2 = m_2 r_2^2 \omega_2$$
$$L_3 = m_3 r_3^2 \omega_3$$

- and

$$L_n = m_n r_n^2 \omega_n$$

- Therefore:

$$L_{net} = m_1 r_1^2 \omega_1 + m_2 r_2^2 \omega_2 + \cdots + m_n r_n^2 \omega_n$$

- Since all points of a rigid body rotate with **same angular velocity**:

$$\omega_1 = \omega_2 = \omega_3 = \omega$$

- Therefore:

$$L_{net} = \omega(m_1 r_1^2 + m_2 r_2^2 + \cdots + m_n r_n^2)$$

- But:

$$I = \sum_{i=1}^{n} m_i r_i^2$$

- Hence:

$$L = I\omega$$

The angular momentum of a rigid body is equal to the product of its moment of inertia and angular velocity.

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Relation b/w Angular Torque & momentum :-

- We know:

$$L = r \times p$$

- Taking change w.r.t time i.e (xplying $\Delta/\Delta t$ on b/s)

$$\frac{\Delta L}{\Delta t} = r \times \frac{\Delta p}{t}$$

$$\frac{\Delta L}{\Delta t} = r \times \frac{\Delta P}{\Delta t}$$

- According to newton's second law of motion in terms of momentum:

$$F = \frac{\Delta P}{\Delta t}$$

- Therefore:

$$\frac{\Delta L}{\Delta t} = r \times F$$

- But torque is defined as:

$$\tau = r \times F$$

- Therefore:

$$\boxed{\frac{\Delta L}{\Delta t} = \tau}$$

oo

$\Delta P = P_f - P_i$

$\Delta P = mv_f - mv_i$

$\Delta P = m(v_f - v_i)$

$\Delta P = m\Delta v$

$\frac{\Delta P}{\Delta t} = \frac{m\Delta v}{\Delta t}$

$\Delta P/\Delta t = ma$

$\Delta P/\Delta t = F$

$F = \frac{\Delta P}{\Delta t}$

"Torque is the rate of change of angular momentum"

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Conservation of angular momentum:-

Principle:

In the absence of any external torque, the angular momentum of a system remains constant.

- Mathematically:

$$\frac{\Delta L}{\Delta t} = 0$$

- Therefore:

$$\Delta L = 0$$

- Since:

$$\Delta L = L_f - L_i$$

- we get:

$$L_f - L_i = 0$$

- Hence:

$$L_f = L_i$$

- Using:

$$L = I\omega$$

- we obtain:

$$I_f\omega_f = I_i\omega_i$$

If no external torque acts on a system, its final angular momentum is equal to its initial angular momentum.

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→ Spinning Ice Skater —o

—o Applications:—

- A spinning ice stater demonstrates conservation of angular momentum.

— When arms are extended:-
  - Moment of interia I is large
  - Angular velocity ω is small.

— When arms are pulled close:-
  - Moment of inertia I becomes smaller.
  - Angular velocity ω becomes larger

- Because:-

$$I\omega = \text{constant}$$

- Therefore:

$$I\downarrow \Rightarrow \omega\uparrow$$

- and:

$$I\uparrow \Rightarrow \omega\downarrow$$

When the skater pulls the arms inward, the mass moves closer to the axis. Hence the moment of inertia decreases. To keep angular momentum constant, the ω increases, so the skater spins faster.

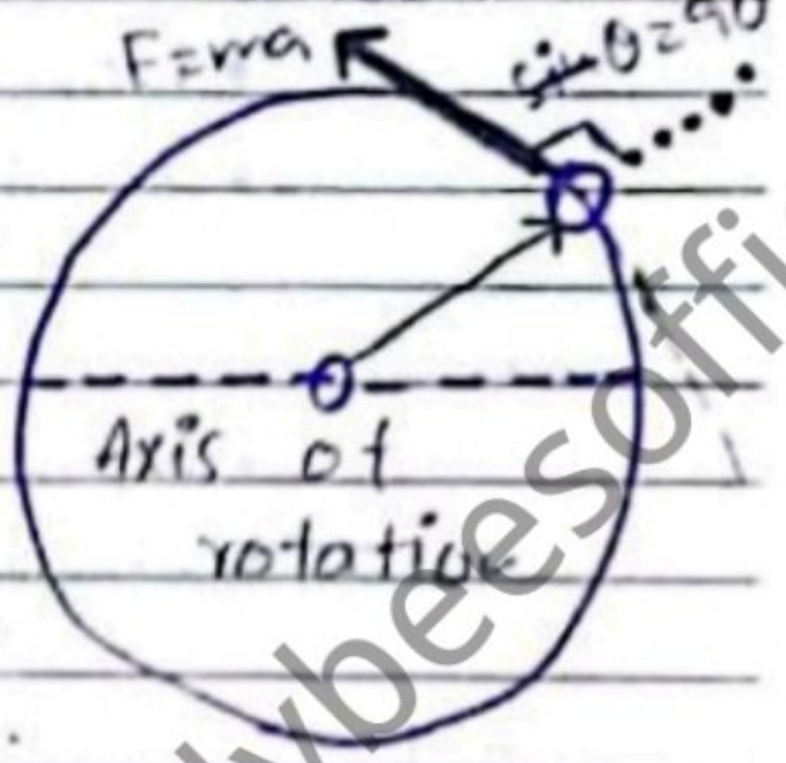
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Torque and Angular accelerations:

Basic Idea:

- There is a relationship b/w torque and angular acceleration, just as Newton's second law relates force and linear acceleration



Torque for a point mass:

- Consider a point mass m rotating at a distance r from the axis of rotation.
- The tangential force F produces torque.

Definition of Torque:

$$\tau = rF\sin\theta$$

- where:
  - τ = torque
  - r = perpendicular distance from axis.
  - F = applied force
  - θ = angle b/w r and F.

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- For tangential force:-
  $\theta = 90°$
- Since:
  $\sin 90° = 1$
- Therefore:
  $\boxed{T = rF}$

Derivation of $T = I\omega$:-

- According to newton's second law:
  $F = ma$
- For rotational motion, tangential acceleration is related to angular acceleration by:
  $\boxed{a = r\alpha}$
- Substituting:
  $F = m(r\alpha)$
  $F = mr\alpha$
- Since:
  $T = rF$
- Therefore:
  $T = r(mr\alpha)$
  $\boxed{T = mr^2\alpha}$
- For a point mass:
  $T = I\alpha$ $\boxed{I = mr^2}$

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- Therefore: $\tau = I\alpha$

❖ **Important Concept:**

$\tau = I\alpha$ is the rotational analogue of Newton's second law.

$F = ma$

COMPARISON

Linear motion:	Rotational motion:
o Force F	o Torque τ
o Mass m	o Moment of Inertia I
o Acceleration a	o Angular acceleration α
o $F = ma$	o $\tau = I\alpha$

Artificial Gravity :-

❖ **What is Artificial Gravity?**

o In a spacecraft or space station, astronauts experience weightlessness because they are not supported in the usual way by a surface.

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- Artificial gravity can be produced by rotating a space station about its own axis.
- When the station rotates, the surface exerts a force on the occupants. This force provides the required centripetal force, causing them to move in a circular path.
- The resulting effect is produced by the occupants as **artificial gravity.**

Centripetal Acceleration:

Suppose the radius of the rotating space station is R and its linear speed is v.

The centripetal acceleration is:

$$a_c = \frac{v^2}{R}$$

To produce an effect similar to Earth's gravity, we require:

$$a_c = g$$

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Therefore:

$$\frac{v^2}{R} = g$$

After rearranging:

$$v^2 = Rg$$

Hence:

$$v = \sqrt{Rg}$$

Angular Velocity of the Space Station:

- The relationship b/w linear and angular velocity is:

$$v = \omega R$$

- From:

$$v = \sqrt{gR}$$

- we get:

$$\sqrt{gR} = \omega R$$

- Therefore:

$$\omega = \sqrt{\frac{g}{R}}$$

- Important Observation:

$$\omega \propto \frac{1}{\sqrt{R}}$$

- So, for the same artificial gravity:
  larger radius → smaller ω

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Time Period of the Satellite :-

"Time period is the time required to complete one rotation."

- Distance travelled in one rotation:

$$2\pi R$$

- Therefore:

$$T = \frac{2\pi R}{v}$$

- Since:

$$v = \omega R$$

- then:

$$T = \frac{2\pi R}{\omega R}$$

- Hence:

$$T = \frac{2\pi}{\omega}$$

- Using:

$$\omega = \sqrt{\frac{g}{R}}$$

- we obtain:

$$\boxed{T = 2\pi \sqrt{\frac{R}{g}}}$$

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Frequency of the Satellite :-

"Frequency is the reciprocal of time period"

$$f = \frac{1}{T}$$

- Since:

$$T = 2\pi \sqrt{\frac{R}{g}}$$

- Therefore:

$$\boxed{f = \frac{1}{2\pi} \sqrt{\frac{g}{R}}}$$

- SI unit: H_z

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